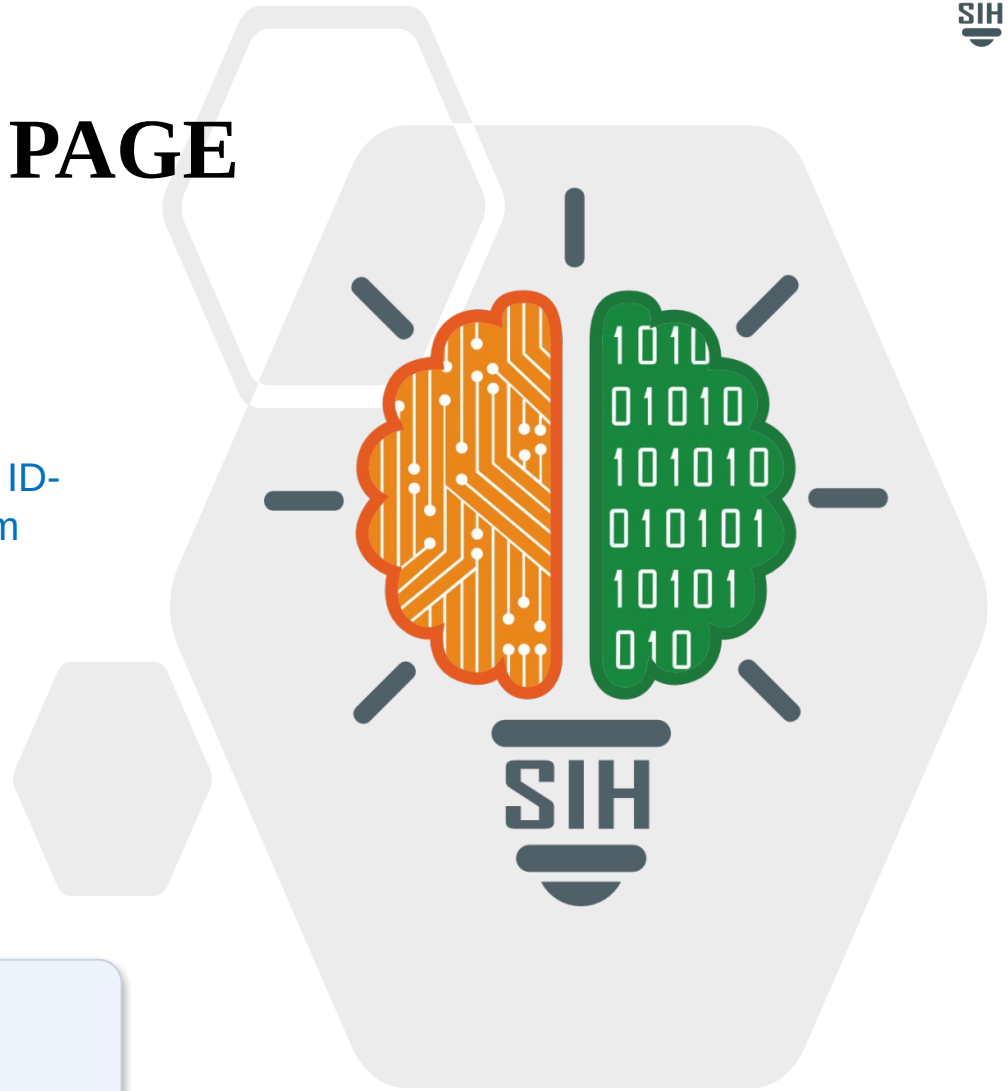
SMART INDIA HACKATHON 2026

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TITLE PAGE



- **Problem Statement ID –** SIH26263
- **Problem Statement Title-** Development of an APAAR ID-Based Student Data Management and Verification Platform for AICTE-Approved Institutions
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal)** AI Riders

VIDYA PRAMAAN

One student. One APAAR ID. One verifiable record.

Working prototype: https://vidya-pramaan-sih26263.onrender.com

AI Riders

VIDYA PRAMAAN

29.6 lakh enrolments in AICTE institutions (2025-26) · **24.25 crore** APAAR IDs issued (Mar 2026)

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❖ Proposed Solution (Describe your Idea/Solution/Prototype)

❖ Detailed explanation of the proposed solution



Institution upload
Excel · CSV · ERP API

APAAR validation
registry + consent

Data verification
31 explained rules

Central repository
one record per student

Analytics & monitoring
after every submission

Authorised verification
consent + signed QR

What is new APAAR, NAD / DigiLocker and ABC hold IDs and awards; none checks each admission against approved intake. We add that check (slide 6).

❖ How it addresses the problem

✖ Only total enrolment reaches AICTE (except PGDM), per the PS	✔ Student-level records keyed on APAAR, per upload
✖ Excess admission: fine of 5× the fee per student (APH)	✔ Seat #61 of 60 refused at upload and at submit
✖ Same student enrolled at two colleges goes unseen	✔ Cross-college match + shared-contact rings
✖ Records are hard to verify at source (PS)	✔ Signed QR, checked even offline; consent for more

❖ Innovation and uniqueness of the solution

Twin-safe matching
Mohd = Mohammed, but Rohit ≠ Mohit

Ghost-admission map
1 phone, 8 students, 2 colleges: flagged

Decisions kept
reason required; never re-flagged

Signed checkpoints
even a full history rewrite is caught

Rules as data
each value sourced; unconfirmed: warn only

APAAR on degrees
university print feed with verification QR

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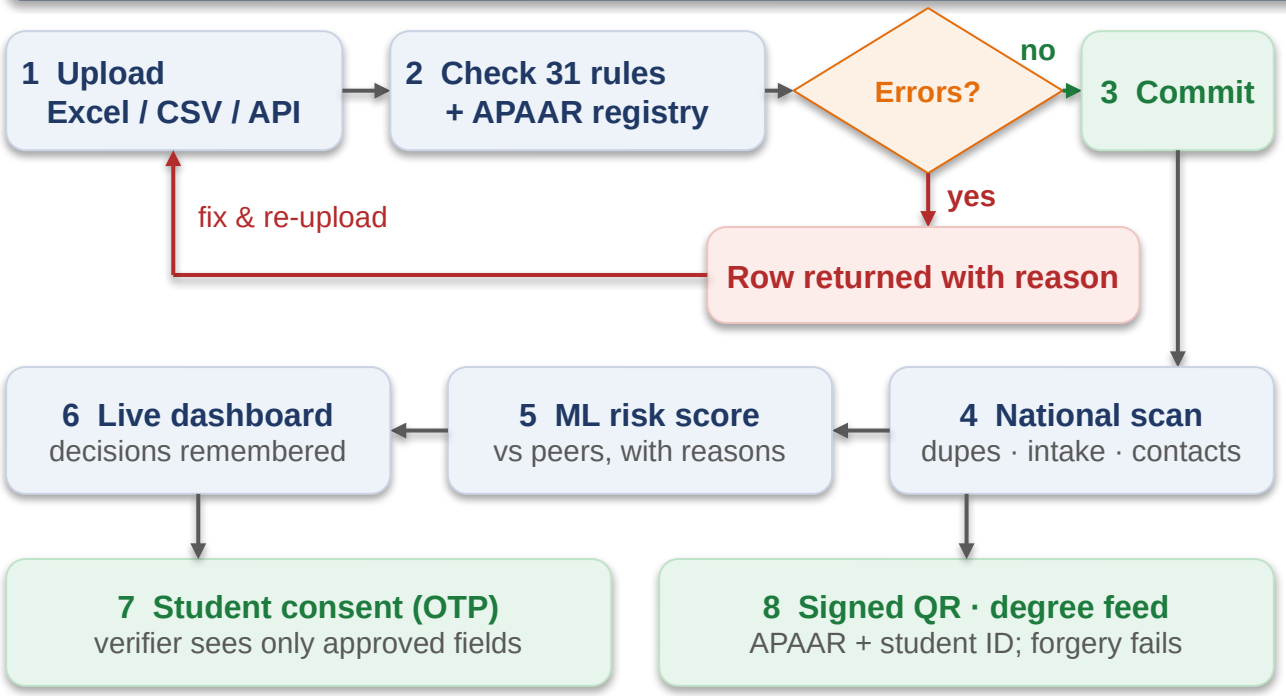
TECHNICAL APPROACH

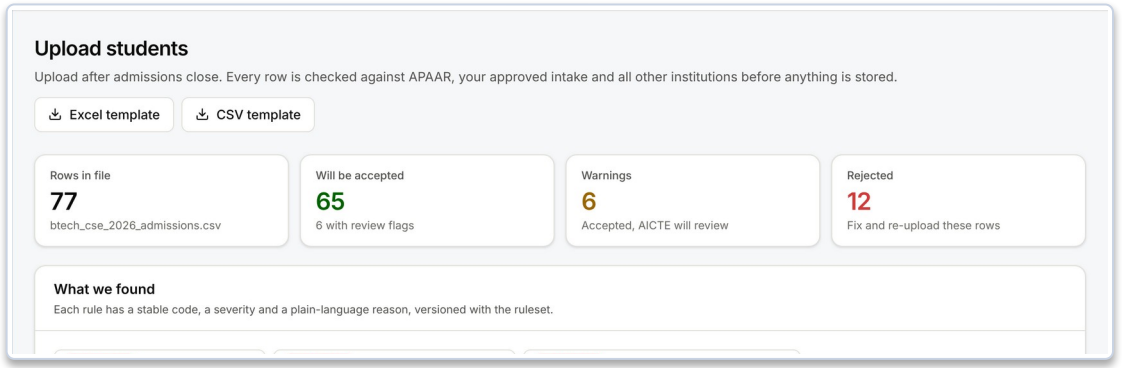
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❖ Technologies to be used (e.g. programming languages, frameworks, hardware)

- **Python · FastAPI** — REST · OpenAPI
- **React · TypeScript** — installable PWA
- **PostgreSQL 16** — SQLite in the demo
- **scikit-learn** — Isolation Forest
- **Name matcher** — twin-safe · Indic
- **Ed25519 · W3C** — did:web · AES-GCM

❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)





Working prototype: 77-row upload → 65 accepted, 12 returned with rule codes

Built: upload, 31 rules, scan, risk score, consent, signed QR, audit

Simulated (no public API yet): APAAR lookup, DigiLocker, portal sign-in

Planned: API Setu / DigiLocker partner access, govt SSO, STQC audit

Try it live: https://vidya-pramaan-sih26263.onrender.com
7 demo roles · synthetic data · first load can take a minute (free server)

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FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea

0.79 / 0.92
held-out precision / recall (old matcher 0.65 / 0.68)

0/80 · 2/12
twins flagged: seeded · held-out (sent for review)

17/17
planted error types; 0 false flags on 56 clean rows

164 · 57 · 4
planted dual enrolments, identity dupes, contact rings

42/46
top-ranked colleges were planted anomalies

65 tests
93% coverage; security and WCAG checks in CI

All synthetic data: regression checks, not real-world accuracy, which the pilot will measure. A flag never acts on its own: an officer decides, with a recorded reason.

Potential challenges and risks	Strategies for overcoming these challenges
No public APAAR API	✔ Registry adapter: mock now → DigiLocker / API Setu after partner onboarding
Students without APAAR (opt-out, FN / OCI)	✔ Provisional ID, never rejected; mother's or guardian's name accepted
Student privacy (DPDP Act; APAAR is voluntary)	✔ Consent per field, contacts encrypted at rest, no Aadhaar stored
Wrong flags: twins, spelling variants, bad data	✔ Rejected rows never reach risk scores; reasons shown; officer decides
Rules change; some values not yet confirmed	✔ Rules file per year; unconfirmed values warn, never reject
ERP formats, low digital readiness	✔ ERP headers auto-mapped, Excel template, API keys, Hindi, phone app

Roadmap M1–3 pilot, 1 state, 50 colleges: measure hours saved and false flags → M4–6 API Setu, SSO, STQC → M7–12 national **Cost** no licence fees

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IMPACT AND BENEFITS

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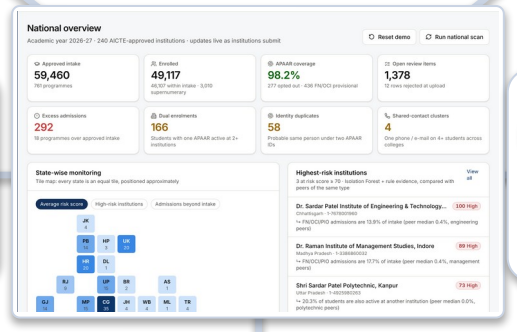
❖ Potential impact on the target audience

AICTE & regulators
excess & dual enrolment caught at submission

Institutions
one upload; errors and corrections explained

Universities
APAAR + student ID printed on degrees

Employers
QR check on a phone; consent for more



One national view

Students
own record; control over who sees it

Today
✖ ₹23.1 cr fine = 42 excess seats × ₹11 lakh fee × 5 (FORE v AICTE, SC 2019)

✖ AISHE 2022-23 data published in 2026

✖ “a number of students have mismatch in Aadhaar and APAAR ID details” (CU Srinagar, 2026)

With Vidya Pramaan
✔ Seat #61 of 60 refused at upload, the same day

✔ National view refreshed after each submission

✔ Mismatch returned with its reason; fixed by re-upload

❖ Benefits of the solution (social, economic, environmental, etc.)

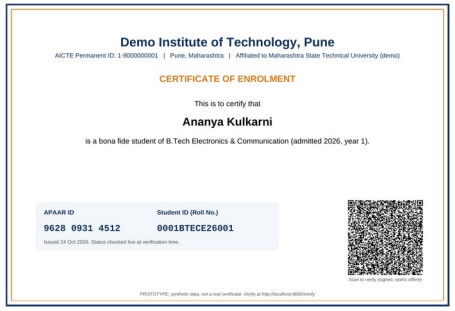
Social
APAAR opt-outs never refused, as the Supreme Court requires

Economic
no licence fees; one upload serves AICTE and university

Environmental
fewer paper returns and verification letters

Governance
audited evidence pack for review and show-cause

Expected benefits; the pilot measures hours saved and flags confirmed.


Demo Institute of Technology, Pune

AICTE Permanent ID: 1-9000000001 | Pune, Maharashtra | Affiliated to Maharashtra State Technical University (demo)

CERTIFICATE OF ENROLMENT

This is to certify that

Ananya Kulkarni

is a bona fide student of B.Tech Electronics & Communication (admitted 2026, year 1).

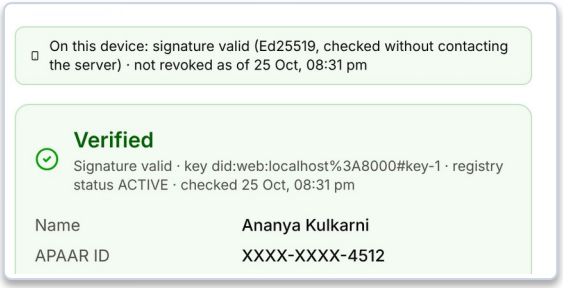
APAAR ID
9628 0931 4512

Student ID (Roll No.)
0001BTECE26001

Issued 24 Oct 2026. Status checked live at verification time.

PROTOTYPE: synthetic data, not a real certificate. Verify at http://localhost:8000/verify

Certificate: APAAR ID + student ID + QR



QR: on-device check + revocation list

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RESEARCH AND REFERENCES

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❖ Details / Links of the reference and research work

Why not just use the systems that exist?

	NAD / DigiLocker	ABC	AISHE	Vidya Pramaan
Student-level admissions vs approved intake	—	—	totals, 2–3 yrs late	✔ per upload
Same student at two colleges; ghost admissions	—	—	—	✔
Supernumerary caps by programme and year	—	—	—	✔
Consent-based verification for employers	✔	—	—	✔ field-level
APAAR + student ID feed for degree printing	issuer uploads	credits only	—	✔

✔ = built and covered by a test (docs/CLAIMS.md); other columns from public documentation (refs 4–5). Vidya Pramaan reuses APAAR and feeds NAD / ABC and AISHE: no new ID, no parallel credential store, no student rejected for opting out.

References

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3. Supreme Court, FORE School of Management v AICTE (2019): 5× fee per excess student sci.gov.in
4. NAD on DigiLocker: consent-based verification (FAQ) nad.digilocker.gov.in
5. AISHE 2022-23 & 2023-24 reports (released 2026) aishe.gov.in
6. APAAR opt-out: Orissa HC, Rohit Anand Das (Dec 2025); Supreme Court direction (Jul 2026) scconline.com · livelaw.in
7. Parliament replies: AICTE enrolment 2025-26 (Lok Sabha); APAAR IDs at 30 Mar 2026 (Rajya Sabha) sansad.in
8. Digital Personal Data Protection Act 2023; DPDP Rules 2025 meity.gov.in
9. W3C Verifiable Credentials and did:web; RFC 8032 (Ed25519) w3.org · rfc-editor.org
10. Liu, Ting & Zhou (2008) Isolation Forest; IDinsight on matching Indian names doi.org/10.1109/ICDM.2008.17 · idinsight.org

Live prototype
https://vidya-pramaan-sih26263.onrender.com
Synthetic data only · 7 demo roles · works on phones

Source code & tests
https://github.com/Pulkitkh/VidyaPramaan
65 tests · 93% coverage · MIT licence